

The five-vertex tournament $C_3[\text{TT}_2, \text{TT}_2, 1]$ is a counterexample to Conjecture 24 of Bonamy, Groenland, Johnston, Morrison and Scott

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Machine-generated note

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Abstract

Bonamy, Groenland, Johnston, Morrison and Scott proved that for every finite graph H other than a clique or an independent set there is a countably infinite H -free graph every locally finite perturbation of which contains an induced copy of H , and they conjectured the tournament analogue (Conjecture 24 of arXiv:2506.08810): for every finite non-transitive tournament T there is a countably infinite T -free tournament S_T every locally finite perturbation of which contains T . We show that the five-vertex tournament $T_* = C_3[\text{TT}_2, \text{TT}_2, 1]$, a directed triangle two of whose vertices are replaced by transitive tournaments on two vertices, is a counterexample. The reason is structural: every countable T_* -free tournament is obtained from a transitive tournament by reversing a locally finite set of arcs, and transitive tournaments are themselves T_* -free; hence every countably infinite T_* -free tournament has a nonempty locally finite perturbation that is still T_* -free. The single-arc variant, Problem 23 of the same paper, is not addressed.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 4](#) records the full provenance.

1 Introduction

Tournaments. A *tournament* is an orientation of a complete graph; we write $x \rightarrow y$ when the arc between x and y is directed from x to y , and $O(x) = N^+(x)$ for the set of out-neighbours of x . For sets X, Y of vertices, $X \rightarrow Y$ means $x \rightarrow y$ for all $x \in X, y \in Y$. A tournament is *transitive* if it contains no directed triangle; equivalently its arcs are $x \rightarrow y$ for $x < y$ in some linear order of its vertex set. TT_n denotes the transitive tournament on n vertices and C_3 the directed triangle. For tournaments T_0, T_1, T_2 , the *composition* $C_3[T_0, T_1, T_2]$ is the tournament on the disjoint union $V(T_0) \cup V(T_1) \cup V(T_2)$ that induces T_i on $V(T_i)$ and has $V(T_0) \rightarrow V(T_1) \rightarrow V(T_2) \rightarrow V(T_0)$; a single vertex is written 1. A tournament S is *T -free* if no set of $|V(T)|$ vertices of S induces a tournament isomorphic to T . Every subtournament of a tournament is induced, so “induced copy of T ” and “copy of T ” coincide for tournaments. A tournament is *countable* if its vertex set is finite or countably infinite; the conjecture below concerns countably infinite ones.

Perturbations. For tournaments S, S' on the same vertex set, the *change graph* $D(S, S')$ is the undirected graph on $V(S)$ whose edges are the pairs whose orientation differs between S and S' . Bonamy, Groenland, Johnston, Morrison and Scott [1, Section 3] define the operative notion for graphs as follows:

For a pair of distinct vertices $u, v \in V(G)$, perturbing the pair uv is the operation which removes uv from $E(G)$ if it is present and adds it if it is not present. A locally finite perturbation of G is a graph obtained from G by perturbing arbitrarily many pairs (at least one, possibly infinitely many) under the constraint that for any vertex v , the number of perturbed pairs involving v is finite.

Their Introduction phrases the same notion as “perturbing a nonempty set of pairs such that every vertex is in a finite number of perturbed pairs; equivalently, we can consider graphs G' such that $E(G) \triangle E(G')$ is the edge set of a locally finite graph” [1, Section 1], which is the change-graph formulation used below. In their concluding section they add that “for tournaments the natural operation is to reverse the direction of edges” [1, Section 8]. Accordingly, a *locally finite perturbation* of a tournament S is a tournament S' on $V(S)$ such that $D(S, S')$ is nonempty and locally finite (every vertex lies in finitely many changed pairs). We stress that nonemptiness is part of the paper’s definition (“at least one”), not a convention introduced here; the paper itself notes that “the locally finite perturbations enforce that S_T is infinite”.

The problem. The main result of [1] (their Theorem 1) is that for every finite graph H which is not a clique or an independent set there is a countably infinite H -free graph G_H such that every locally finite perturbation of G_H has an induced copy of H . Section 8 of [1] asks for analogues in other structures and poses, for tournaments, first the single-arc problem and then the conjecture we refute. Both are quoted verbatim.

Problem 1.1 ([1, Problem 23]). For which finite tournaments H , does there exist a countable tournament G that does not contain H as subtournament, yet any tournament obtained from G by changing the direction of a single arc does contain H ?

Conjecture 1.2 ([1, Conjecture 24]). Let T be a finite tournament. If T is not transitive, then there exists a countably infinite T -free tournament S_T such that every locally finite perturbation of S_T has an induced copy of T .

The conjecture is introduced with the sentence “Perhaps the following analogue of Theorem 1 is true for the ‘locally finite perturbation’ variant of this question.” Its only hypothesis is that T is not transitive, and the paper explains why this hypothesis is necessary: every tournament on 2^n vertices contains a transitive subtournament on n vertices, so a transitive T cannot be avoided by an infinite tournament at all. There is no restriction on the order, connectivity or modular structure of T . The catalog entry for this problem (id 2506.08810__03) and the referee’s own searches found no published work addressing [Conjecture 1.2](#); the referee also found that the journal version (Canadian Journal of Mathematics, published online 24 March 2026) contains no retraction of, or note on, the tournament conjecture.

The counterexample. Let $T_* = C_3[\text{TT}_2, \text{TT}_2, 1]$ be the tournament on the five vertices a_0, a_1, b_0, b_1, c with

$$a_0 \rightarrow a_1, \quad b_0 \rightarrow b_1, \quad \{a_0, a_1\} \rightarrow \{b_0, b_1\} \rightarrow \{c\} \rightarrow \{a_0, a_1\},$$

that is, with the ten arcs

$$\begin{aligned} a_0 \rightarrow a_1, \quad a_0 \rightarrow b_0, \quad a_0 \rightarrow b_1, \quad a_1 \rightarrow b_0, \quad a_1 \rightarrow b_1, \\ b_0 \rightarrow b_1, \quad b_0 \rightarrow c, \quad b_1 \rightarrow c, \quad c \rightarrow a_0, \quad c \rightarrow a_1. \end{aligned}$$

The out-degrees of (a_0, a_1, b_0, b_1, c) are $(3, 2, 2, 1, 2)$. T_* has exactly four directed triangles, namely $c \rightarrow a_i \rightarrow b_j \rightarrow c$ for $i, j \in \{0, 1\}$, all through c ; in particular it is not transitive, and the two triangles $c \rightarrow a_0 \rightarrow b_0 \rightarrow c$ and $c \rightarrow a_1 \rightarrow b_1 \rightarrow c$ share no arc. Both $\{a_0, a_1\}$ and $\{b_0, b_1\}$ are nontrivial modules, so T_* is not prime; T_* is strongly connected but $T_* - c$ is not; and the minimum number of arcs whose reversal makes T_* transitive is 2.

Theorem 1.3 (Main theorem). *Let S be a countably infinite T_* -free tournament. Then there is a T_* -free tournament $S' \neq S$ on $V(S)$ such that the change graph $D(S, S')$ is locally finite. Equivalently: every countably infinite T_* -free tournament has a locally finite perturbation (in the sense of [1]) that is still T_* -free.*

Corollary 1.4. *Conjecture 1.2 is false. The tournament $T = T_*$ is finite and not transitive, yet no countably infinite T_* -free tournament S_T has the property that every locally finite perturbation of S_T contains an induced copy of T_* .*

Proof. T_* is a finite tournament containing a directed triangle, so it satisfies the only hypothesis of [Conjecture 1.2](#). Let S_T be any countably infinite T_* -free tournament. [Theorem 1.3](#) supplies $S' \neq S_T$ on the same vertex set with $D(S_T, S')$ locally finite; since $S' \neq S_T$ the set of reversed pairs is nonempty, so S' is a locally finite perturbation of S_T in the paper's sense, and S' is T_* -free, i.e. it has no induced copy of T_* . Hence S_T does not have the conjectured property. $\square$

Idea of proof. Two finite facts about T_* drive the argument, and the referee checked both by exhaustive enumeration. First, if an arc $x \rightarrow y$ of a T_* -free tournament admits a vertex p with $x \rightarrow p \rightarrow y$, then at most two vertices z satisfy $y \rightarrow z \rightarrow x$ ([Lemma 2.2](#)): any three such z together with x, y, p contain a copy of T_* . So an arc lying in infinitely many directed triangles has no such p , and arcs with no such p form an undirected graph of maximum degree at most 2 ([Lemma 2.3](#)). Hence the graph $J(S)$ of arcs in infinitely many directed triangles has maximum degree at most 2, and a general criterion ([Lemma 2.1](#)) shows that a countable tournament with locally finite $J(S)$ differs from some transitive tournament L by a locally finite set of reversals ([Theorem 2.4](#)). If S is not transitive then $L \neq S$, and L is T_* -free because T_* is not transitive. Second, if S is transitive, reversing a single arc creates only directed triangles through that arc, whereas T_* contains two arc-disjoint directed triangles; so the result is still T_* -free ([Lemma 2.5](#)). [\[full proof\]](#)

2 Structure of T_* -free tournaments

For an arc $x \rightarrow y$ of a tournament S put

$$Z(x, y) = \{z : y \rightarrow z \rightarrow x\}, \quad P(x, y) = \{p : x \rightarrow p \rightarrow y\}.$$

$Z(x, y)$ is the set of vertices completing $x \rightarrow y$ to a directed triangle, and $P(x, y)$ the set of vertices that would do so if the arc xy were reversed. Note that $O(y) \setminus O(x) = Z(x, y)$: indeed $x \notin O(y)$ because $x \rightarrow y$, so a vertex $z \in O(y) \setminus O(x)$ is distinct from x and y , satisfies $y \rightarrow z$ and, not being an out-neighbour of x , satisfies $z \rightarrow x$. Let $J(S)$ be the undirected graph on $V(S)$ in which xy is an edge precisely when the arc between x and y lies in infinitely many directed triangles, i.e. when $Z(x, y)$ (for $x \rightarrow y$) is infinite.

Lemma 2.1 (Criterion for locally finite distance from a transitive tournament). *Let S be a countable tournament. There is a transitive tournament L on $V(S)$ with $D(S, L)$ locally finite if and only if $J(S)$ is locally finite.*

Idea of proof. Necessity: if $xy \notin E(D(S, L))$ then every directed triangle xyz of S must have xz or yz in $D(S, L)$, else it survives into L ; so $J(S) \subseteq D(S, L)$. Sufficiency: call $x \sim y$ if $O(x) \triangle O(y)$ is finite. On the classes, “ $O(y) \setminus O(x)$ finite” defines a partial order; take a linear extension (Szpilrajn [2]; for countably many classes an explicit insertion argument suffices). By $O(y) \setminus O(x) = Z(x, y)$, any arc between different classes that points against this order lies in $J(S)$. Inside a class C , fix $r \in C$ and split C into the in-neighbours C^- and out-neighbours C^+ of r ; finiteness of $O(x) \triangle O(r)$ gives finite out-degrees inside C^- , finite in-degrees inside C^+ , and only locally finitely many arcs from C^+ to C^- . Ordering C^- in reverse enumeration order, then r , then C^+ in enumeration order, makes the discrepancies inside C locally finite. Only the sufficiency direction is used below. [\[full proof\]](#)

Lemma 2.2. *Let S be T_* -free and let $x \rightarrow y$ be an arc with $P(x, y) \neq \emptyset$. Then $|Z(x, y)| \leq 2$.*

Idea of proof. Take $p \in P(x, y)$ and suppose $z_1, z_2, z_3 \in Z(x, y)$. All five vertices are distinct, and two of the z_i , say z_1, z_2 , are oriented the same way relative to p . If $p \rightarrow z_1, z_2$ then $\{p, y\} \rightarrow \{z_1, z_2\} \rightarrow$

$\{x\} \rightarrow \{p, y\}$ with $p \rightarrow y$; if $z_1, z_2 \rightarrow p$ then $\{z_1, z_2\} \rightarrow \{x, p\} \rightarrow \{y\} \rightarrow \{z_1, z_2\}$ with $x \rightarrow p$. Either way the five vertices induce $C_3[\text{TT}_2, \text{TT}_2, 1] = T_*$ (a two-vertex block is TT_2 whichever way its arc points). [\[full proof\]](#)

Lemma 2.3. *In any tournament, the arcs $x \rightarrow y$ with $P(x, y) = \emptyset$ form an undirected graph of maximum degree at most 2: every vertex has at most one such out-arc and at most one such in-arc.*

Idea of proof. If $x \rightarrow y$ and $x \rightarrow z$ both had empty P , then $y \rightarrow z$ would put $y \in P(x, z)$ and $z \rightarrow y$ would put $z \in P(x, y)$. In-arcs are symmetric. [\[full proof\]](#)

Theorem 2.4 (Structure theorem). *Let S be a T_* -free tournament. Then $J(S)$ has maximum degree at most 2. Consequently, if S is countable, there is a transitive tournament L on $V(S)$ such that $D(S, L)$ is locally finite: every countable T_* -free tournament is obtained from a transitive tournament by reversing a locally finite set of arcs.*

Idea of proof. An edge xy of $J(S)$ with $x \rightarrow y$ has $Z(x, y)$ infinite, hence of size at least 3, so $P(x, y) = \emptyset$ by Lemma 2.2. Thus $J(S)$ is a subgraph of the graph of Lemma 2.3, and Lemma 2.1 applies. [\[full proof\]](#)

Lemma 2.5. *Let L be a transitive tournament and let L' be obtained from L by reversing a single arc. Then L' is T_* -free.*

Idea of proof. Every directed triangle of L' contains the reversed arc, since otherwise it would be a directed triangle of L . But T_* contains two arc-disjoint directed triangles, so no copy of T_* fits into L' . [\[full proof\]](#)

The structure theorem is one-directional: a locally finite perturbation of a transitive tournament may well contain T_* . Indeed T_* itself is obtained from TT_5 with the order $a_0 < a_1 < b_0 < b_1 < c$ by reversing the two arcs a_0c and a_1c , and no single reversal suffices (Lemma 2.5), so the inversion number of T_* is exactly 2. What Theorem 2.4 says is that nothing outside the locally finite perturbations of transitive tournaments is T_* -free.

3 Remarks

Remark 3.1 (Interpretation audit). Because the object is tiny, the substance of the referee’s audit was to check that Conjecture 1.2 as posed really admits it. The referee fetched Section 8 of arXiv:2506.08810v3 and found the conjecture’s statement to agree character for character with the one attacked (Conjecture 1.2 above); it fetched the definition of “locally finite perturbation” quoted in the Introduction and found the writeup’s reading, a nonempty set of reversed arcs with every vertex in finitely many of them, to be the paper’s own. Three consequences were recorded. (i) The nonemptiness caveat the writeup states is not an assumption it imposed but the paper’s convention (“at least one”); had the writeup instead used the empty perturbation, the disproof would be a strawman, and it does not. This is why the case of a transitive S in Theorem 1.3 needs the separate Lemma 2.5: the transitive tournament L produced by Theorem 2.4 coincides with S there, so a genuine reversal must be exhibited. (ii) “Induced copy” means “subtournament”, so there is no induced-versus-weak ambiguity to exploit. (iii) The conjecture has no order threshold and no primeness or connectivity hypothesis; its only hypothesis, non-transitivity, is satisfied by T_* , which has four directed triangles. The referee concluded that a reasonable author of Conjecture 1.2 would accept the argument as a refutation of the conjecture as posed.

Remark 3.2 (Why the conjecture admits a five-vertex counterexample). The referee was instructed to explain why five authors would have missed a five-vertex counterexample, and gave three answers, which we reproduce as its assessment. (i) Conjecture 1.2 is a speculation in the open-problems section of a paper about graphs, introduced with “Perhaps the following analogue of Theorem 1 is true”; nothing

in the paper indicates a small-tournament check, and the paper’s ancillary code (`small_graphs.py`, `theorem_10.py`) concerns the graph results. (ii) The intuition behind Theorem 1 of [1] is that once the degenerate targets (cliques and independent sets) are excluded, the space of H -free graphs has “points that are isolated in a very strong sense”; the natural transfer replaces the degenerate targets by transitive tournaments. For T_* this fails for a tournament-specific reason: the obstruction is not that T_* is hard to hit but that it is so easy to avoid that, by Theorem 2.4, the whole space of countable T_* -free tournaments collapses onto the locally finite perturbations of transitive tournaments; and transitive tournaments are themselves T_* -free, so this space has no isolated point at all. (iii) The machinery of [1] replaces vertices by infinite cliques or independent sets; the tournament analogue replaces vertices by infinite transitive blocks. But $T_* = C_3[TT_2, TT_2, 1]$ is exactly a directed triangle with two vertices blown up into transitive blocks of size 2, so any blow-up of a directed triangle with two blocks of size at least 2 already contains T_* , and the technique cannot get started for this target. The referee also measured how special T_* is: among all tournaments on 4, 5 and 6 vertices up to isomorphism (4, 12 and 56 of them), T_* is the *only* one for which both finite engines of the proof work, namely all 64 completions of the configuration in Lemma 2.2 contain T , and T has two arc-disjoint directed triangles. The counterexample is thus a narrow object rather than one that should have been obvious.

Remark 3.3 (Scope: Problem 23 is untouched). Theorem 1.3 says nothing about Problem 1.1 for T_* . In the non-transitive case its perturbation L in general reverses infinitely many arcs, and in the transitive case Lemma 2.5 shows only that a transitive tournament is not a solution to Problem 1.1 for $H = T_*$. Whether some countable T_* -free tournament becomes non- T_* -free under every single arc reversal remains open, as the writeup itself states. Likewise the writeup proposes no repaired conjecture; the question for which non-transitive T the conclusion of Conjecture 1.2 holds is open. The earlier attempt discussed in Remark 3.4 proves it for several classes.

Remark 3.4 (Relation to the earlier attempt). This target was a second attempt. The first, by gpt-5.6-sol (attacks/2506.08810__03/output.md, verdict “partial”, unverified), proved the conclusion of Conjecture 1.2 for every tournament one arc reversal away from transitive, for the cyclic tournament C_5 via a C_3 -lexicographic construction, and for tournaments that stay strongly connected after deleting any two vertices and have no proper transitive module (including C_7), and it named low-connectivity tournaments with nontrivial modular structure as the untreated region. The present argument reuses nothing from it. The referee checked that the two are consistent: T_* needs 2 reversals to become transitive (exhaustive over all 120 orderings), has the nontrivial modules $\{a_0, a_1\}$ and $\{b_0, b_1\}$, and $T_* - c$ is not strongly connected; so none of the earlier positive results covers T_* , which falls exactly into the region that attempt flagged as open. Had the inversion number of T_* been 1, one of the two documents would be wrong.

Remark 3.5 (Computational checks by the referee). The referee’s scripts (pure Python, in the repository folder `verification_astra/scripts/2506.08810__03/`) established the following; none failed.

- *The object.* T_* is a tournament with out-degree sequence $(3, 2, 2, 1, 2)$ on (a_0, a_1, b_0, b_1, c) ,¹ four directed triangles all through c , a pair of arc-disjoint directed triangles, empty intersection of the arc sets of all four triangles, inversion number 2, nontrivial modules $\{a_0, a_1\}$ and $\{b_0, b_1\}$, strongly connected with $T_* - c$ not strongly connected.
- *Lemma 2.2 is a complete enumeration.* The configuration $x \rightarrow y, p \in P(x, y), z_1, z_2 \in Z(x, y)$ with z_1, z_2 alike relative to p has 4 completions, all isomorphic to T_* ; the six-vertex configuration with $z_1, z_2, z_3 \in Z(x, y)$ has $2^3 \cdot 2^3 = 64$ completions and 0 of them are T_* -free.
- *Lemma 2.3.* Three-vertex configurations with two P -empty out-arcs at a vertex: 0; with two P -empty in-arcs: 0.
- *Finite shadow of Theorem 2.4.* Over all labelled tournaments on n vertices, the arcs lying in at least three directed triangles of a T_* -free tournament always form a graph of maximum degree

¹The referee’s report lists this as $(3, 2, 2, 2, 1)$ in the same order; b_1 has out-degree 1 (it beats only c) and c has out-degree 2, so the report’s two last entries are transposed. The multiset, and everything built on it, is unaffected. We re-derived all invariants in this paragraph.

at most 2 with $P = \emptyset$ throughout:

n	labelled tournaments	T_* -free	max. degree of the ≥ 3 -triangle graph	such an arc with $P \neq \emptyset$
4	64	64	0	none
5	1 024	904	1	none
6	32 768	17 648	1	none
7	2 097 152	434 352	2	none

The bound 2 is attained at $n = 7$, so the constant in [Lemma 2.3](#) is tight. A further 24 T_* -free tournaments on 8, 9, 10 and 12 vertices found by randomised local search satisfy both conclusions.

- [Lemma 2.5](#). Reversing each of the $\binom{n}{2}$ arcs of TT_n for $n = 5, \dots, 9$ (110 cases) creates 0 copies of T_* .
- *Infinite examples*. On finite windows of eight explicit countable tournaments, the four that contain T_* (among them $C_3[\text{TT}, \text{TT}, \text{TT}]$, the circular tournament on $\mathbb{Z}_{17}$, and the earlier attempt’s C_3 -lexicographic power) are as [Theorem 2.4](#) requires, and the four T_* -free ones ($\text{TT}_\omega[C_3]$; TT_ω with the pairs $(2t, 2t+1)$ reversed; TT_ω with all pairs $(i, i+1)$ reversed; evens dominating odds minus a matching) have triangle graph of maximum degree at most 1 on the window. For these four the recipe of [Lemma 2.1](#) was worked out by hand and its change graph measured on windows of 20, 60, 120 and 240 vertices: the maximum degree stays at 1, 1, 2 and 1 respectively, constant in the window size. The referee notes that “ $O(x) \triangle O(y)$ finite” has no faithful finite-window proxy, which is why this check was done by hand rather than automatically.
- *Uniqueness on at most six vertices*, as described in [Remark 3.2](#).

Remark 3.6 (Use of choice). The sufficiency direction of [Lemma 2.1](#) orders the $\sim$ -classes by a linear extension of a partial order and enumerates each class. In general the existence of a linear extension is Szpilrajn’s theorem [2]; for a countable tournament there are countably many classes and the extension can be built by inserting the classes one at a time ([Appendix A](#)), so the argument uses no choice beyond fixing enumerations of a countable set. The writeup did not name Szpilrajn; the referee called this cosmetic.

4 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The counterexample and its proof were found by the language model gpt-6-astra (reasoning effort max, flex tier) in a single autonomous pass started on 2026-09-17, in the retry leg of the campaign (catalog id 2506.08810__03, leg attacks_retry; writeup in attacks_retry/2506.08810__03/output.md). This was a second attempt at the problem: the prompt included the earlier attempt by gpt-5.6-so1 (attacks/2506.08810__03/output.md, verdict “partial”, no referee report), which is summarised in [Remark 3.4](#). The new writeup states that it uses no result of the earlier one, and the referee confirmed that nothing is inherited; the partial results of [Remark 3.4](#) are the earlier model’s and are not claimed here. (2) An independent adversarial referee agent (claude-opus-5) reviewed the writeup on 2026-09-17. It fetched arXiv:2506.08810v3 and checked the statement of Conjecture 24 and the definition of “locally finite perturbation” verbatim against the writeup’s reading, re-derived every step (in particular the three “it follows that” bullets in the sufficiency direction of [Lemma 2.1](#) and the implicit step that a cross-class arc contradicting the linear extension lies in $J(S)$), verified [Lemma 2.2](#), [Lemma 2.3](#) and [Lemma 2.5](#) by complete enumeration, checked the degree bound over all 2 097 152 labelled tournaments on 7 vertices, checked consistency with the earlier attempt, and searched for prior or concurrent work, finding none. Its verdict was CONFIRMED, high confidence. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1; the compilation was repaired and the quotations from the paper re-checked on 2026-09-18. The deviations from the writeup are the following, and no other mathematical content was changed.

- The text was reorganised with full proofs in [Appendix A](#). The writeup’s Corollary 4 is stated as [Theorem 2.4](#), and its “Case 2” is isolated as [Lemma 2.5](#).
- The explicit arc list of T_* and its invariants (out-degrees, directed triangles, modules, connectivity, inversion number) were added; they come from the referee’s step 1 and were recomputed for this note, correcting the labelling slip noted in [Remark 3.5](#).
- In [Lemma 2.1](#): the existence of a linear extension is attributed to Szpilrajn [2] and an explicit construction for countably many classes is supplied (referee caveat 3); the step that an arc between classes pointing against the linear extension, and not merely against the partial order, lies in $J(S)$ is spelled out (referee step S4); the three bullets on C^- and C^+ are derived in full (referee step S5); the well-definedness and transitivity of the order on classes are written out.
- In [Lemma 2.2](#) the distinctness of the five vertices is made explicit (referee step S7).
- The verbatim quotations of the paper’s definition of locally finite perturbation (from its Section 3 and, in the equivalent change-graph form, its Section 1), of Problem 23, of the sentence introducing Conjecture 24 and of the remark that the perturbations force S_T to be infinite were added. The referee fetched Conjecture 24 and the definition from arXiv:2506.08810v3; all five passages were re-fetched from the same source and confirmed character for character on 2026-09-18. The referee report locates the definition in Section 2 of the paper; it is in Section 3 (“Notation and definitions”), and the note cites it there.
- The paragraph following [Theorem 2.4](#), observing that T_* is TT_5 with the two arcs a_0c and a_1c reversed and hence has inversion number exactly 2, was added; the value 2 is the referee’s exhaustive computation, and the explicit pair of arcs was checked by hand against the arc list in the Introduction.
- The bibliographic data of the journal version (DOI 10.4153/S0008414X26102132, published online 24 March 2026) were taken from the publisher’s page on 2026-09-18.
- [Remarks 3.1](#), [3.2](#) and [3.4](#) to [3.6](#) are taken from the referee report; they contain no claim of the writeup’s.

No human mathematician has checked this argument. We circulate it for human review, not as a claim of a settled result.

A Full proofs

Throughout, for an arc $x \rightarrow y$ of the tournament under consideration, $Z(x, y) = \{z : y \rightarrow z \rightarrow x\}$, $P(x, y) = \{p : x \rightarrow p \rightarrow y\}$, $O(x) = N^+(x)$, and $X \subseteq^* Y$ means that $X \setminus Y$ is finite. We use repeatedly the identity

$$O(y) \setminus O(x) = Z(x, y) \quad \text{whenever } x \rightarrow y, \quad (1)$$

proved at the start of [Section 2](#): $x \notin O(y)$ since $x \rightarrow y$, so $z \in O(y) \setminus O(x)$ means $z \neq x, y, y \rightarrow z$ and $z \rightarrow x$.

Lemma 2.1 (Criterion for locally finite distance from a transitive tournament). *Let S be a countable tournament. There is a transitive tournament L on $V(S)$ with $D(S, L)$ locally finite if and only if $J(S)$ is locally finite.*

Proof. Necessity. Let L be transitive on $V(S)$ with $D = D(S, L)$ locally finite. We show $J(S) \subseteq D$, which gives local finiteness of $J(S)$. Let $xy \notin E(D)$, so xy has the same orientation in S and L , and let z be such that x, y, z span a directed triangle of S . If neither xz nor yz were in $E(D)$, all three arcs of this triangle would have the same orientation in L , and L would contain a directed triangle, contradicting transitivity. Hence $z \in N_D(x) \cup N_D(y)$, a finite set. So xy lies in only finitely many directed triangles of S , i.e. $xy \notin E(J(S))$.

Sufficiency. Assume $J(S)$ is locally finite.

Step 1: classes. Define $x \sim y$ iff $O(x) \triangle O(y)$ is finite. This is reflexive and symmetric, and transitive because $O(x) \triangle O(z) \subseteq (O(x) \triangle O(y)) \cup (O(y) \triangle O(z))$. Let $\mathcal{C}$ be the set of $\sim$ -classes; it is countable.

Step 2: a partial order on classes. For distinct classes C, D declare $C \prec D$ if $O(y) \subseteq^* O(x)$ for some (equivalently every) $x \in C, y \in D$. This is independent of the representatives: if $x' \sim x$ and $y' \sim y$ then $O(y') \setminus O(x') \subseteq (O(y') \setminus O(y)) \cup (O(y) \setminus O(x)) \cup (O(x) \setminus O(x'))$, and the first and third sets are finite. It is transitive, since $O(z) \setminus O(x) \subseteq (O(z) \setminus O(y)) \cup (O(y) \setminus O(x))$. It is antisymmetric: if $C \prec D$ and $D \prec C$ then both $O(y) \setminus O(x)$ and $O(x) \setminus O(y)$ are finite, so $x \sim y$ and $C = D$. Thus $\prec$ is a strict partial order on $\mathcal{C}$.

Step 3: a linear extension. Let $<$ be a linear order on $\mathcal{C}$ extending $\prec$. Such an order exists by Szpilrajn's theorem [2]; since $\mathcal{C}$ is countable we can also construct it directly. Enumerate $\mathcal{C} = \{C_1, C_2, \dots\}$ and build linear orders $<_n$ on $\{C_1, \dots, C_n\}$ extending $\prec$, each extending the previous one. Given $<_n$, let $\mathcal{P} = \{C_i : i \leq n, C_i \prec C_{n+1}\}$ and $\mathcal{Q} = \{C_j : j \leq n, C_{n+1} \prec C_j\}$. For $C_i \in \mathcal{P}$ and $C_j \in \mathcal{Q}$ transitivity gives $C_i \prec C_j$, hence $C_i <_n C_j$; so every element of $\mathcal{P}$ is $<_n$ -below every element of $\mathcal{Q}$. Insert C_{n+1} immediately after the $<_n$ -largest element of $\mathcal{P}$ (at the beginning if $\mathcal{P} = \emptyset$). The result $<_{n+1}$ is a linear order placing C_{n+1} above $\mathcal{P}$ and below $\mathcal{Q}$, hence extending $\prec$. The union $<$ of the $<_n$ is a linear order on $\mathcal{C}$ extending $\prec$.

Step 4: arcs between classes. We claim: if $x \in C, y \in D, C \neq D, C < D$ and $y \rightarrow x$ in S , then $xy \in E(J(S))$. Suppose not: then $Z(y, x)$ is finite, so by (1) (applied to the arc $y \rightarrow x$) $O(x) \setminus O(y)$ is finite, i.e. $O(x) \subseteq^* O(y)$, which by definition means $D \prec C$, hence $D < C$, contradicting $C < D$. So every pair between distinct classes whose orientation in S disagrees with $<$ is an edge of $J(S)$.

Step 5: inside a class. Fix a class C and $r \in C$, and put $C^- = \{x \in C : x \rightarrow r\}$, $C^+ = \{x \in C : r \rightarrow x\}$, so $C = C^- \cup \{r\} \cup C^+$. For $x \in C$ the set $F(x) = O(x) \triangle O(r)$ is finite. We verify three facts.

- (i) If $x, y \in C^-$ and $x \rightarrow y$, then $y \in O(x)$ and $y \notin O(r)$ (as $y \rightarrow r$), so $y \in O(x) \setminus O(r) \subseteq F(x)$. Hence every $x \in C^-$ has finitely many out-neighbours in C^- .
- (ii) If $x, y \in C^+$ and $y \rightarrow x$, then $y \in O(r)$ and $y \notin O(x)$, so $y \in O(r) \setminus O(x) \subseteq F(x)$. Hence every $x \in C^+$ has finitely many in-neighbours in C^+ .
- (iii) If $x \in C^-, y \in C^+$ and $y \rightarrow x$, then $y \in O(r) \setminus O(x) \subseteq F(x)$ and $x \in O(y) \setminus O(r) \subseteq F(y)$ (as $x \rightarrow r, x \notin O(r)$). Hence the bipartite graph of pairs $x \in C^-, y \in C^+$ with $y \rightarrow x$ has all degrees finite.

Enumerate $C^- = \{u_1, u_2, \dots\}$ and $C^+ = \{v_1, v_2, \dots\}$ (finite or infinite enumerations, without repetition) and order C as

$$\dots < u_3 < u_2 < u_1 < r < v_1 < v_2 < v_3 < \dots, \quad (2)$$

i.e. C^- in reverse enumeration order, then r , then C^+ in enumeration order. Let L_C be the transitive tournament on C given by this order ($a \rightarrow b$ iff $a < b$). We count the pairs inside C on which S and L_C disagree at a fixed vertex. At u_i : with r none, since $u_i \rightarrow r$ in both; with u_j for $j < i$ at most $i - 1$ pairs; with u_j for $j > i$, L_C has $u_j \rightarrow u_i$, so a disagreement means $u_i \rightarrow u_j$ in S , and by (i) there are finitely many such j ; with v_j , L_C has $u_i \rightarrow v_j$, so a disagreement means $v_j \rightarrow u_i$ in S , finitely many by (iii). At v_i : with r none; with v_j for $j < i$ at most $i - 1$; with v_j for $j > i$, L_C has $v_i \rightarrow v_j$, so a disagreement means $v_j \rightarrow v_i$ in S , finitely many by (ii); with u_j finitely many by (iii). At r : none. So $D(S|_C, L_C)$ is locally finite.

Step 6: assembling L . Order $V(S)$ by: x before y if the class of x is $<$ the class of y , or if they lie in the same class C and x precedes y in (2). This is a linear order on $V(S)$; let L be the corresponding transitive tournament. A vertex $x \in C$ lies in finitely many pairs of $D(S, L)$: those inside C are finite in number by Step 5, and those with vertices of other classes are edges of $J(S)$ at x by Step 4, of which there are finitely many because $J(S)$ is locally finite. Hence $D(S, L)$ is locally finite. $\square$

Lemma 2.2. *Let S be T_* -free and let $x \rightarrow y$ be an arc with $P(x, y) \neq \emptyset$. Then $|Z(x, y)| \leq 2$.*

Proof. Let $p \in P(x, y)$, so $x \rightarrow p \rightarrow y$, and suppose for a contradiction that $Z(x, y)$ contains three distinct vertices z_1, z_2, z_3 , so $y \rightarrow z_i \rightarrow x$. The vertices x, y, p, z_1, z_2, z_3 are pairwise distinct: $p \neq x, y$ and $z_i \neq x, y$ by definition, and $p \neq z_i$ because $p \rightarrow y$ while $y \rightarrow z_i$. By the pigeonhole principle two of the z_i , say z_1 and z_2 , are oriented the same way relative to p .

Case $p \rightarrow z_1$ and $p \rightarrow z_2$. Put $A = \{p, y\}$, $B = \{z_1, z_2\}$ and $c = x$. Then $p \rightarrow y$ inside A , so A induces TT_2 ; B induces TT_2 whichever way the arc between z_1 and z_2 points; $A \rightarrow B$ because $p \rightarrow z_1, z_2$ (the case hypothesis) and $y \rightarrow z_1, z_2$ (as $z_i \in Z(x, y)$); $B \rightarrow \{c\}$ because $z_i \rightarrow x$; and $\{c\} \rightarrow A$ because $x \rightarrow p$ and $x \rightarrow y$. So $\{x, y, p, z_1, z_2\}$ induces $C_3[\text{TT}_2, \text{TT}_2, 1] = T_*$.

Case $z_1 \rightarrow p$ and $z_2 \rightarrow p$. Put $A = \{z_1, z_2\}$, $B = \{x, p\}$ and $c = y$. Then A induces TT_2 ; $x \rightarrow p$ inside B ; $A \rightarrow B$ because $z_i \rightarrow x$ and $z_i \rightarrow p$; $B \rightarrow \{c\}$ because $x \rightarrow y$ and $p \rightarrow y$; and $\{c\} \rightarrow A$ because $y \rightarrow z_i$. Again $\{x, y, p, z_1, z_2\}$ induces T_* .

Both cases contradict the T_* -freeness of S . Hence $|Z(x, y)| \leq 2$. $\square$

Lemma 2.3. *In any tournament, the arcs $x \rightarrow y$ with $P(x, y) = \emptyset$ form an undirected graph of maximum degree at most 2: every vertex has at most one such out-arc and at most one such in-arc.*

Proof. Let x be a vertex and suppose $x \rightarrow y$ and $x \rightarrow z$ with $y \neq z$ both satisfy $P(x, y) = P(x, z) = \emptyset$. Either $y \rightarrow z$, in which case $x \rightarrow y \rightarrow z$ gives $y \in P(x, z)$, or $z \rightarrow y$, in which case $z \in P(x, y)$; both are impossible. So at most one out-arc of x has empty P . Dually, suppose $y \rightarrow x$ and $z \rightarrow x$ with $y \neq z$ both satisfy $P(y, x) = P(z, x) = \emptyset$. Either $y \rightarrow z$, giving $y \rightarrow z \rightarrow x$ and $z \in P(y, x)$, or $z \rightarrow y$, giving $y \in P(z, x)$; both are impossible. So at most one in-arc of x has empty P , and x has degree at most 2 in the undirected graph formed by the arcs with empty P . $\square$

Theorem 2.4 (Structure theorem). *Let S be a T_* -free tournament. Then $J(S)$ has maximum degree at most 2. Consequently, if S is countable, there is a transitive tournament L on $V(S)$ such that $D(S, L)$ is locally finite: every countable T_* -free tournament is obtained from a transitive tournament by reversing a locally finite set of arcs.*

Proof. Let $xy \in E(J(S))$ with $x \rightarrow y$. Then $x \rightarrow y$ lies in infinitely many directed triangles, i.e. $Z(x, y)$ is infinite and in particular $|Z(x, y)| \geq 3$. By Lemma 2.2, $P(x, y) = \emptyset$. Hence every edge of $J(S)$ is an arc with empty P , and Lemma 2.3 bounds the degree of every vertex in $J(S)$ by 2. In particular $J(S)$ is locally finite, and if S is countable the sufficiency direction of Lemma 2.1 yields a transitive tournament L on $V(S)$ with $D(S, L)$ locally finite. $\square$

Lemma 2.5. *Let L be a transitive tournament and let L' be obtained from L by reversing a single arc. Then L' is T_* -free.*

Proof. Let $u \rightarrow v$ be the reversed arc, so L' has $v \rightarrow u$ and agrees with L on every other pair. If a directed triangle of L' did not use the arc $v \rightarrow u$, all three of its arcs would be arcs of L , and L would contain a directed triangle, which is impossible. Hence every directed triangle of L' contains the arc $v \rightarrow u$, and any two directed triangles of L' share an arc. Now suppose $X \subseteq V(L')$ induces a copy of T_* . The triangles $c \rightarrow a_0 \rightarrow b_0 \rightarrow c$ and $c \rightarrow a_1 \rightarrow b_1 \rightarrow c$ of T_* use the arc sets $\{ca_0, a_0b_0, b_0c\}$ and $\{ca_1, a_1b_1, b_1c\}$, which are disjoint; their images are two arc-disjoint directed triangles of L' , a contradiction. So L' is T_* -free. $\square$

Theorem 1.3 (Main theorem). *Let S be a countably infinite T_* -free tournament. Then there is a T_* -free tournament $S' \neq S$ on $V(S)$ such that the change graph $D(S, S')$ is locally finite. Equivalently: every countably infinite T_* -free tournament has a locally finite perturbation (in the sense of [1]) that is still T_* -free.*

Proof. Let S be a countably infinite T_* -free tournament.

Case 1: S is not transitive. By [Theorem 2.4](#) there is a transitive tournament L on $V(S)$ with $D(S, L)$ locally finite. Since S is not transitive and L is, $L \neq S$. Every subtournament of a transitive tournament is transitive, and T_* is not, so L is T_* -free. Take $S' = L$.

Case 2: S is transitive. As S has infinitely many vertices it has an arc $u \rightarrow v$; let S' be obtained from S by reversing it. Then $S' \neq S$, $D(S, S')$ is the single edge uv , which is locally finite, and S' is T_* -free by [Lemma 2.5](#).

In both cases $S' \neq S$ is a T_* -free tournament on $V(S)$ with $D(S, S')$ locally finite. Since $S' \neq S$, the set of reversed pairs is nonempty, so S' is a locally finite perturbation of S in the sense of [1], and it is T_* -free. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step numbers (S1–S11) and lemma numbers refer to that writeup, not to this note: the writeup’s Lemma 1, 2, 3 and Corollary 4 are [Lemmas 2.1 to 2.3](#) and [Theorem 2.4](#) here, and its §4 Cases 1 and 2 are the proof of [Theorem 1.3](#) and [Lemma 2.5](#). The scripts it mentions are in `verification_astra/scripts/2506.08810__03/`. The out-degree tuple in its step S1 has its last two entries transposed (see [Remark 3.5](#)). Treat it as machine output, not as an authority.

B.1 Interpretation

Conjecture as the source states it. I pulled the LaTeXML rendering of arXiv:2506.08810v3 (<https://arxiv.org/html/2506.08810v3>) and extracted Section 8. Conjecture 24 reads, verbatim:

Conjecture 24. Let T be a finite tournament. If T is not transitive, then there exists a countably infinite T -free tournament S_T such that every locally finite perturbation of S_T has an induced copy of T .

This is character-for-character the statement in `prompt.md`. The paper’s definition of the operative term, also verbatim (Section 2, “Locally finite perturbation”):

For a pair of distinct vertices $u, v \in V(G)$, perturbing the pair uv is the operation which removes uv from $E(G)$ if it is present and adds it if it is not present. A locally finite perturbation of G is a graph obtained from G by perturbing **arbitrarily many pairs (at least one, possibly infinitely many)** under the constraint that for any vertex v , the number of perturbed pairs involving v is finite.

and, in the sentence introducing Problem 23, “For tournaments the natural operation is to reverse the direction of edges”.

So the intended tournament reading is exactly: a nonempty set of reversed arcs, each vertex in finitely many reversed pairs. **This is precisely the writeup’s reading.** Three consequences:

1. The writeup’s caveat “perturbations are understood to be nonempty” is not an assumption it imposed — it is the paper’s own convention (“at least one”). Good; had the writeup instead exploited the *empty* perturbation it would have been a strawman, and it does not.
2. “Induced copy of T ” in a tournament is just “subtournament”, so there is no induced/weak ambiguity to game.
3. The conjecture has no order threshold, no primeness hypothesis, no connectivity hypothesis and no “sufficiently large T ” clause. A five-vertex nontransitive T is squarely in scope. I looked for any such hidden hypothesis and there is none; the only hypothesis is “ T is not transitive”, and T_* is not transitive (verified, step 1).

The instructed question — why would the authors have missed a five-vertex counterexample? Three answers, all of which I think are the real ones rather than an artifact:

- Conjecture 24 is a *speculation in the concluding open-problems section of a paper about graphs*, introduced with the words “Perhaps the following analogue of Theorem 1 is true for the ‘locally finite perturbation’ variant of this question.” It is not presented as a tested conjecture, and nothing in the paper suggests the authors ran any small-tournament check. (The paper does ship ancillary code, `small_graphs.py` and `theorem_10.py`, but both are for the *graph* results.)
- The natural transfer of intuition from Theorem 1 is “the only obstruction is the degenerate one”, i.e. cliques/independent sets become transitive tournaments. That guess is wrong for a reason that is specific to tournaments: the obstruction is not that T_* is hard to *hit*, it is that T_* is so easy to *avoid* that the whole space of countable T_* -free tournaments collapses onto locally-finite perturbations of the transitive order — and the transitive order is itself T_* -free, so the space has no isolated point at all.
- Their machinery blows vertices up into infinite cliques / infinite independent sets. The tournament analogue is blowing up into infinite *transitive* blocks — and $T_* = C_3[TT_2, TT_2, 1]$ is built so that a transitive block of size 2 sitting inside a C_3 already is a copy of T_* . The technique cannot get off the ground for this particular target.

A reasonable author of Conjecture 24 would, I believe, accept this as a genuine refutation of the conjecture as posed (not of Problem 23, the single-arc variant, which the writeup explicitly does not claim and which remains open).

Retry-specific check. The prompt carried an earlier `gpt-5.6-sol` attempt claiming `partial` (the conjecture holds for inversion-number-one tournaments, for C_5 , and for prime 3-vertex-strong tournaments). The new writeup inherits nothing from it — correctly, since nothing is reusable — and I checked that the two are **not in contradiction**, which is a real consistency hazard here: $\iota(T_*) = 2$ (computed exhaustively over all 120 orderings), T_* has the nontrivial modules $\{a_0, a_1\}$ and $\{b_0, b_1\}$ so it is not prime, and $T_* - \{c\}$ is not strongly connected so it is not 3-vertex-strong. Hence none of the earlier attempt’s positive results covers T_* . It falls exactly into the region that attempt flagged as untreated in its own Section 8 (“low-connectivity tournaments with nontrivial modular structure”). Had $\iota(T_*)$ been 1, one of the two documents would have to be wrong; it is 2.

B.2 Step-by-step findings

I number the writeup’s steps as S1–S9.

step	label	note
S1. Definition of $T_* = C_3[TT_2, TT_2, 1]$ (§1)	VALID	$A = \{a_0, a_1\}, B = \{b_0, b_1\}, \{c\}$, $a_0 \rightarrow a_1, b_0 \rightarrow b_1, A \rightarrow B \rightarrow c \rightarrow A$. Machine-built: it is a tournament, outdegree sequence $(3, 2, 2, 2, 1)$, nontransitive, strongly connected, exactly 4 directed triangles, all through c .
S2. Identity $O(y) \setminus O(x) = Z(x, y)$ for $x \rightarrow y$ (eq. (1), §2)	VALID	$x \notin O(y)$ since $x \rightarrow y$, so $O(y) \setminus O(x) = \{z \neq x, y : y \rightarrow z, z \rightarrow x\} = Z(x, y)$. Re-derived; no off-by-one (the writeup does not silently include x or y).
S3. Lemma 1, necessity (S near-transitive $\Rightarrow J(S)$ locally finite)	VALID	If $xy \notin D$ then any directed triangle xyz of S has $xz \in D$ or $yz \in D$ (else it survives into the transitive L), so at most $\deg_D(x) + \deg_D(y) < \infty$ such z . Hence $J(S) \subseteq D$. Correct — but this direction is never used ; only sufficiency is load-bearing.
S4. Lemma 1, sufficiency : class construction ($\sim$, the $\subseteq^*$ poset, statement (2))	VALID	$x \sim y \iff O(x) \triangle O(y)$ finite is an equivalence relation (transitivity via $\triangle_{AC} \subseteq \triangle_{AB} \cup \triangle_{BC}$). Precedence “ C precedes D iff $O(y) \subseteq^* O(x)$ ” is representative-independent, transitive, and antisymmetric (both directions $\Rightarrow$ finite $\triangle \Rightarrow$ same class). A linear extension exists (Szpilrajn; for countable V one can do it by hand). If $x \rightarrow y$ and $xy \notin J(S)$ then $Z(x, y)$ is finite, so by (1) $O(y) \subseteq^* O(x)$, so $\text{cl}(x)$ precedes $\text{cl}(y)$; contrapositive gives (2) exactly. I also checked the converse direction the writeup leaves implicit: if $x \in C, y \in D, C$ before D in the extension but $y \rightarrow x$ in S , then $yx \notin J(S)$ would put D before C in the <i>poset</i> and hence in the extension — contradiction. So (2) is right.
S5. Lemma 1, sufficiency : the three bullets about C^-, C^+	VALID (terse but I re-derived all three)	This is the one “it follows that” in the paper, so I checked each. Fix $r \in C$; for $x \in C, O(x) \triangle O(r)$ is finite. (i) $x \in C^-, y \in C^-$, $x \rightarrow y$: $y \in O(x)$ and $y \notin O(r)$ (as $y \rightarrow r$), so $y \in O(x) \setminus O(r)$, finite — finite outdegree in C^- . (ii) $x \in C^+, y \in C^+$, $y \rightarrow x$: $y \in O(r) \setminus O(x)$, finite — finite indegree in C^+ . (iii) $x \in C^-, y \in C^+$, $y \rightarrow x$: from x ’s side $y \in O(r) \setminus O(x)$, from y ’s side $x \in O(y) \setminus O(r)$; both finite, so the disagreeing cross graph is locally finite. All three correct.

step	label	note
S6. Lemma 1, sufficiency : the order $C^- < r < C^+$ and local finiteness (eq. (3))	VALID	With C^- in reverse enumeration order, L orients $v_j \rightarrow v_i$ for $j > i$; a discrepancy at v_i with a later-enumerated v_j means $v_i \rightarrow v_j$, i.e. v_j is in v_i 's (finite) out-neighbourhood in C^- ; with an earlier-enumerated v_j there are at most $i - 1$. Dual argument for C^+ with enumeration order and finite indegree. No discrepancies with r (all of $C^- \rightarrow r \rightarrow C^+$ matches). Concatenating with the class order yields a genuine linear order, hence a transitive L , and each vertex sees finitely many changed pairs: within-class (finite, by the above) plus cross-class (inside $J(S)$, locally finite). Correct.
S7. Lemma 2 (S T_* -free, $P(x, y) \neq \emptyset \Rightarrow Z(x, y) \leq 2$)	VALID — and machine-verified exhaustively	All five vertices x, y, p, z_1, z_2 are forced distinct ($p \rightarrow y$ vs $y \rightarrow z_i$ rules out $p = z_i$). The only free orientations are p vs the z_i (pigeonhole gives two alike) and $z_1 z_2$. I enumerated all resulting configurations and every one is isomorphic to T_* (see computational check). The block identifications the writeup asserts are right: case $p \rightarrow z_i$ gives $A = \{p, y\}$ (with $p \rightarrow y$), $B = \{z_1, z_2\}$, $c = x$; case $z_i \rightarrow p$ gives $A = \{z_1, z_2\}$, $B = \{x, p\}$ (with $x \rightarrow p$), $c = y$.
S8. Lemma 3 (arcs with $P(x, y) = \emptyset$ have max degree ≤ 2) and Corollary 4	VALID — machine-verified	Two out-arcs at x with $P = \emptyset$: whichever way yz points, one of them acquires a P -vertex. Dually for in-arcs. Exhaustive 3-vertex check: 0 violating configurations either way. Corollary 4: an arc in infinitely many directed triangles has $ Z = \infty \geq 3$, so Lemma 2 forces $P = \emptyset$, so $J(S)$ is a subgraph of Lemma 3's graph, so $\Delta(J(S)) \leq 2$; Lemma 1 then applies. Chain is airtight.
S9. §4 Case 1 (S nontransitive)	VALID	Corollary 4 gives a transitive L on $V(S)$ at locally finite distance; $L \neq S$ because S is nontransitive, so the set of reversed pairs is nonempty (= a perturbation in the paper's sense); L is T_* -free because T_* is nontransitive and transitive tournaments have only transitive subtournaments.
S10. §4 Case 2 (S transitive)	VALID — machine-verified	Reversing one arc uv of a transitive S : any directed triangle of S' avoiding $\{u, v\}$ would be a directed triangle of S , impossible; so every directed triangle of S' uses the arc $v \rightarrow u$. T_* has two arc-disjoint directed triangles ($c \rightarrow a_0 \rightarrow b_0 \rightarrow c$ and $c \rightarrow a_1 \rightarrow b_1 \rightarrow c$; I confirmed the intersection of <i>all four</i> of T_* 's directed triangles' arc sets is empty), so $T_* \not\subseteq S'$. Verified by brute force on TT_n , $n = 5, \dots, 9$, for every arc.
S11. Conclusion (no S_{T_*} exists)	VALID	Both cases exhaust the countably infinite T_* -free tournaments and each produces a nonempty locally finite perturbation that is still T_* -free. T_* is nontransitive, so Conjecture 24 is false.

No step is labelled GAP or ERROR. The single place where the writeup leans on “it follows that” is S5, and all three of its bullets are one-line consequences that I re-derived in full.

B.3 Reference check

- **arXiv:2506.08810 (v3), Conjecture 24.** Fetched the full text (<https://arxiv.org/html/2506.08810v3>, LaTeXML render) and located Section 8. Conjecture 24 is quoted above and matches prompt.md exactly. Its only hypothesis is “ T is not transitive”; the accompanying footnote justifies that hypothesis via “any tournament on 2^n elements has a transitive subtournament on n elements”, which is the standard greedy bound and is not used by the writeup. **Confirmed.**

- **arXiv:2506.08810, definition of “locally finite perturbation.”** Fetched and quoted verbatim above; includes “at least one, possibly infinitely many”. **Confirmed**, and it is the writeup’s reading.
- **Problem 23 (single-arc variant).** Present in the same section, distinct from Conjecture 24. The writeup correctly disclaims any bearing on it. **Confirmed**.
- **No external theorem is cited by the writeup.** The proof is entirely self-contained (the only non-elementary ingredient is the existence of a linear extension of a partial order, which the writeup does not name; Szpilrajn’s theorem, or an explicit countable construction, supplies it). There is therefore nothing to misquote — which removes the single most common failure mode.
- **Prior-art / already-known check.** Two web searches for a follow-up disproving Conjecture 24 or producing a tournament counterexample returned nothing: only the source paper (arXiv + the Pith review page), unrelated induced-saturation and tournament papers. The paper itself (published in Canad. J. Math., FirstView 2026) contains no retraction or note about the tournament conjecture. So this is **not** ALREADY_KNOWN as far as I can establish.

B.4 Computational check

Scripts under `verification_astra/scripts/2506.08810__03/` (pure `itertools`, `stdlib`, `python3`; `networkx` is not installed in the repo `venv` and is not needed):

- `tstar.py` — shared encoding of T_* and tournament utilities.
- `step1_object.py` — builds T_* and audits it.
- `step2_lemmas.py` — the lemma enumerations (the heavy one; ~4 min, dominated by the exhaustive $n = 7$ sweep).
- `step3_infinite.py` — the structure theorem against explicit infinite constructions.
- `step4_lemma1_recipe.py` — Lemma 1’s conclusion on explicit infinite T_* -free tournaments.
- `step5_scope.py` — how special T_* is.

1. **The object (step 1).** T_* is a tournament; outdegrees $(a_0, a_1, b_0, b_1, c) = (3, 2, 2, 2, 1)$; **nontransitive**; strongly connected; T_* — $\{c\}$ not strongly connected; exactly 4 directed triangles, all containing c ; **a pair of arc-disjoint directed triangles exists** ($\{a_1 b_1, b_1 c, c a_1\}$ and $\{a_0 b_0, b_0 c, c a_0\}$) and the intersection of all four triangles’ arc sets is **empty**. Inversion number $\iota(T_*) = 2$ (exhaustive over all $5! = 120$ orders). Nontrivial modules: $\{a_0, a_1\}$ and $\{b_0, b_1\}$, so **not prime**. Also, incidentally, T_* is locally transitive.

2. **Lemma 2 (step 2) — this is a complete proof, not a sample.** The lemma’s configuration has only finitely many free orientations, so enumeration settles it:

- the 4 configurations $\{\text{case} \in \{p \rightarrow z_i, z_i \rightarrow p\} \times \{z_1 \rightarrow z_2, z_2 \rightarrow z_1\} \text{ are all isomorphic to } T_*(4/4);$
- the full 6-vertex version $(x \rightarrow y, p \in P(x, y), z_1, z_2, z_3 \in Z(x, y), \text{ all } 2^3 \cdot 2^3 = 64 \text{ orientation choices})$: **0 of the 64 are T_* -free**. So no T_* -free tournament has an arc with $P \neq \emptyset$ and $|Z| \geq 3$.

3. **Lemma 3 (step 2) — also complete.** 3-vertex configurations with two P -empty out-arcs at a vertex: **0**. With two P -empty in-arcs: **0**.

4. **Corollary 4’s finite shadow (step 2).** In every T_* -free tournament the arcs lying in ≥ 3 directed triangles should form a graph of $\max \text{degree} \leq 2$, with $P = \emptyset$ throughout. Exhaustive over all labelled tournaments:

n	labelled tournaments	T_* -free	max degree of the ≥ 3 -triangle graph	any such arc with $P \neq \emptyset$
4	64	64	0	no
5	1024	904	1	no
6	32768	17648	1	no
7	2097152	434352	2	no

Plus 24 T_* -free tournaments on $n = 8, 9, 10, 12$ produced by randomized local search: all satisfy both conclusions. The bound 2 is attained at $n = 7$, i.e. Lemma 3’s constant is tight and not slack.

5. **Case 2 (step 2).** For TT_n , $n = 5, \dots, 9$, reversing each of the $\binom{n}{2}$ arcs: **0** copies of T_* created, in all $10 + 15 + 21 + 28 + 36 = 110$ cases.

6. **The structure theorem against explicit infinite tournaments (step 3).** Windows of eight explicit countable constructions. Those that contain T_* (so are irrelevant as would-be counterexamples to the structure theorem) are: $C_3[TT, TT, TT]$ (copy at vertices 0, 1, 2, 3, 4), the circular local order C_{17} (copy at 0, 1, 2, 3, 10), the C_3 -lexicographic power S_Δ of the *previous* attempt (copy at 0, 1, 3, 4, 6), and “transitive + one vertex beating exactly the evens” (copy at 0, 2, 4, 5, 7) — the last is exactly the hand-worked case where $J(S)$ has infinite degree, and it does contain T_* as Corollary 4 demands. Those that are T_* -free — $TT_\omega[C_3]$, TT with each $(2t, 2t + 1)$ flipped, TT with all $(i, i + 1)$ flipped, and “evens $\rightarrow$ odds minus a matching” — all have $\Delta(J_3) \leq 1$ on the window.

7. **Lemma 1’s conclusion on those four (step 4).** *Caveat on method:* “ $O(x) \triangle O(y)$ finite” has no faithful finite-window proxy — my first attempt used a threshold and it misclassified the classes in one example (both by splitting and, after switching to union-find, by merging), which is an artifact of truncation and not a defect of Lemma 1. So I instead worked out the class decomposition by hand for each example, wrote down the order Lemma 1’s recipe produces, and measured the change graph on growing windows:

construction	order produced by the recipe	max change degree at $n = 20/60/120/240$
$TT_\omega[C_3]$	blocks in order; inside a block $C^- < r < C^+$	1 / 1 / 1 / 1
TT , pairs $(2t, 2t + 1)$ flipped	natural order	1 / 1 / 1 / 1
TT , all $(i, i + 1)$ flipped	natural order	2 / 2 / 2 / 2
evens $\rightarrow$ odds minus a matching	evens increasing, then odds increasing	1 / 1 / 1 / 1

Constant in the window size, i.e. locally finite in the limit, exactly as Lemma 1 asserts.

8. How special is T_* (step 5). For every tournament T on 4, 5 and 6 vertices up to isomorphism (4, 12 and 56 of them), I tested the two finite engines of the disproof: (A) all 64 six-vertex configurations of Lemma 2 contain T , and (B) T has two arc-disjoint directed triangles. T_* is the unique tournament on at most 6 vertices passing both (none at $n = 4$, exactly one at $n = 5 - T_*$ — none at $n = 6$). That is a strong independent signal that the writeup did not stumble on a generic object: it isolated the one small tournament for which this collapse happens.

No computational check failed.

B.5 Caveats

- Nonempty perturbations.** The disproof needs “perturbation $\Rightarrow$ at least one reversed pair”, otherwise Case 1 would be vacuous when S is already transitive. This is not an assumption the writeup smuggled in: it is the paper’s own definition. Correctly handled, and Case 2 exists precisely to cover the transitive S .
- Problem 23 is untouched.** The writeup says so explicitly and it is right: Case 1 in general reverses infinitely many arcs, so nothing here bears on the single-arc question (“is there a T_* -free countable tournament in which reversing any *one* arc creates T_* ?”). A reader skimming only the verdict line could over-read the scope; the writeup’s own caveats field states it.
- Choice.** Lemma 1’s sufficiency uses a linear extension of a partial order (Szpilrajn) and choices of enumerations. For countable $V(S)$ this is eliminable by an explicit construction; the writeup neither names Szpilrajn nor remarks on this. Cosmetic.
- Lemma 1’s necessity direction is dead weight** — correct, but never invoked. Harmless.
- The writeup proposes no repaired conjecture.** It disproves Conjecture 24 as posed but does not discuss what the right hypothesis should be (e.g. restricting to prime tournaments, or to tournaments with no “ P -empty-arc” collapse, or replacing “nontransitive” by something stronger). Given the step-5 finding that T_* is the unique small tournament of its kind, the natural follow-up — “for which T does Conjecture 24 hold?” — is wide open and the writeup does not claim otherwise. This is an expository gap, not a mathematical one.
- Small-case edge behaviour.** Nothing degenerate: T_* has 5 vertices, S is countably infinite by hypothesis so it has arcs, and Lemma 1 is applied only to countable S . There is no $k = 0/1$, empty-graph or disconnectedness edge case in play. Directed triangles are counted unambiguously (a triangle “containing the arc $x \rightarrow y$ ” $\leftrightarrow$ an element of $Z(x, y)$), so there is no off-by-one hiding in the definition of $J(S)$.
- Verdict-vs-proof match.** The one-line verdict says “every countable tournament avoiding it has a distinct locally finite perturbation that still avoids it” — that is exactly the Theorem in §1 and exactly what is proved. No overclaim.

B.6 Referee summary

I set out to find the error and could not. The interpretation is not merely defensible, it is the paper’s own: I pulled Conjecture 24 and the definition of “locally finite perturbation” verbatim from arXiv:2506.08810v3 and both match the writeup’s reading, including the “at least one” clause that the writeup flags as a caveat. The proof has exactly two finite engines and both are completely machine-checkable, so I checked them by exhaustive enumeration rather than by reading: Lemma 2’s configuration has only 4 (resp. 64) possible completions and every single one contains T_* ; Lemma 3’s obstruction has only 2 completions per case and neither occurs; a transitive tournament with one arc reversed is T_* -free for all 110 arcs of $TT_5, \dots, TT_9$; and across all 2 097 152 labelled tournaments on 7 vertices, all 434 352 that are T_* -free satisfy the degree- ≤ 2 conclusion (with the bound attained, so it is tight). The one genuinely infinitary step, Lemma 1’s sufficiency, is the place a single-pass proof would normally break — its “it follows that” bullets about C^- and C^+ , and the reverse-enumeration ordering trick — so I re-derived all of it line by line and it holds, including the point the writeup leaves implicit (that a cross-class pair contradicting the *linear extension*, not just the poset, must lie in $J(S)$). I also probed for the two failure modes specific to a retry target: the new writeup borrows nothing from the earlier gpt-5.6-so1 attempt, and — the real hazard — it does not contradict it, since $\iota(T_*) = 2$, T_* is not prime and is not 3-vertex-strong, so T_* sits exactly in the region that attempt itself flagged as untreated. Finally, the “why did five authors miss a five-vertex counterexample” worry has a satisfying answer on both sides: Conjecture 24 is a one-sentence speculation (“Perhaps the following analogue of Theorem 1 is true”) in the open-problems section of a paper about graphs, with no small-case verification anywhere in it; and on my side, an exhaustive scan of all tournaments on at most 6 vertices shows T_* is the *unique* one for which this argument works, so it is a genuinely narrow object rather than something that should have been obvious. Verdict: CONFIRMED, high confidence. Conjecture 24 of arXiv:2506.08810 is false, with $C_3[TT_2, TT_2, 1]$ as a counterexample; Problem 23 of the same paper remains open.

References

- [1] M. Bonamy, C. Groenland, T. Johnston, N. Morrison and A. Scott, Infinite induced-saturated graphs, [arXiv:2506.08810](https://arxiv.org/abs/2506.08810) (v3, September 2025); Canadian Journal of Mathematics, FirstView (2026), [doi:10.4153/S0008414X26102132](https://doi.org/10.4153/S0008414X26102132), published online 24 March 2026. Problem 23 and Conjecture 24 are in Section 8.
- [2] E. Szpilrajn, Sur l’extension de l’ordre partiel, Fundamenta Mathematicae 16 (1930), 386–389.